

SQL Server Assignment – Contoso Dataset

Topics Covered: SELECT, WHERE, LIKE, DISTINCT, ORDER BY, GROUP BY, HAVING, TOP N, CASE WHEN, IF ELSE, CTEs, Subqueries, JOINS, System Stored Procedures (application only)

Section 1: Basic Queries

1. Retrieve all products with ListPrice > 100.
2. Display distinct product categories using DISTINCT.
3. Find products where ProductName LIKE '%Bike%'.
4. Retrieve top 10 most expensive products using TOP.
5. List customers ordered by CustomerName.

Section 2: Filtering & Sorting

6. Retrieve orders where TotalDue > 1000.
7. Find customers whose email contains 'gmail'.
8. Sort products by StandardCost descending.
9. Retrieve products where Size IS NOT NULL.
10. Display orders between two dates.

Section 3: Aggregations

11. Find total sales per product using GROUP BY.
12. Count number of orders per customer.
13. Find average order value.
14. Retrieve categories having total sales > 5000 using HAVING.
15. Find max and min ListPrice.

Section 4: CASE WHEN

16. Categorize products as 'Expensive' or 'Cheap' based on ListPrice.
17. Classify orders as 'High', 'Medium', 'Low' based on TotalDue.

Section 5: Subqueries

18. Find products with ListPrice greater than average price.
19. Retrieve customers who placed orders above average order value.
20. Find second highest TotalDue using subquery.

Section 6: CTEs

21. Use CTE to calculate total sales per product and filter top 5.
22. Use CTE to rank customers based on spending.

Section 7: IF ELSE

23. Write a query using IF ELSE to check if total orders exceed 100.

24. Use IF ELSE to categorize business performance based on revenue.

Section 8: JOINS

25. Retrieve all orders with corresponding customer names (INNER JOIN).

26. List all products sold in each order (JOIN Products, OrderDetails).

27. Find customers who have not placed any orders (LEFT JOIN).

28. Retrieve total sales per product using joins.

29. Display order details with product name, quantity, and sales amount.

30. Find total revenue generated per customer using joins.

Section 9: System Stored Procedures

31. Use sp_help to view table structure.

32. Use sp_columns to list columns of Products table.

33. Use sp_spaceused to check table size.